Neoplasia

Molecular Basis of Cancer (LG10)

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Learning outlines:

1. Carcinogenesis.
2. Tumor progression.
3. Features of malignant cells.

Carcinogenesis

Carcinogenesis is a multistep process at both the phenotypic and the genetic levels, resulting from the accumulation of multiple mutations.
Malignant neoplasms have several phenotypic attributes, such as excessive growth, local invasiveness, and the ability to form distant metastases.

Ability to Invade and Metastasize

The metastatic cascade can be subdivided into two phases: invasion of ECM and vascular dissemination and homing of tumor cells.

- Invasion of the ECM is an active process that requires four steps :

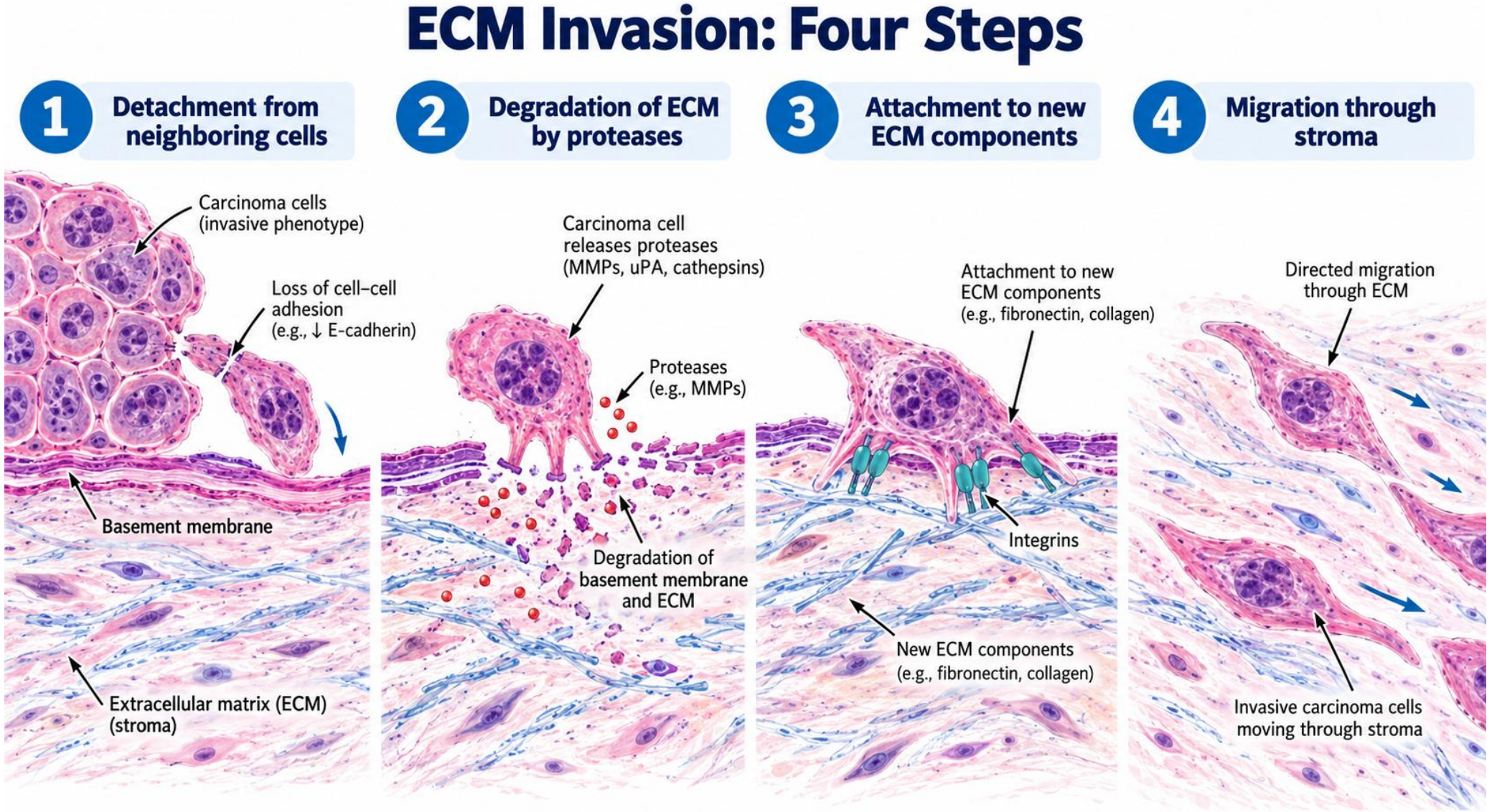
1-Detachment of tumor cells from each other.

2-Degradation of ECM.

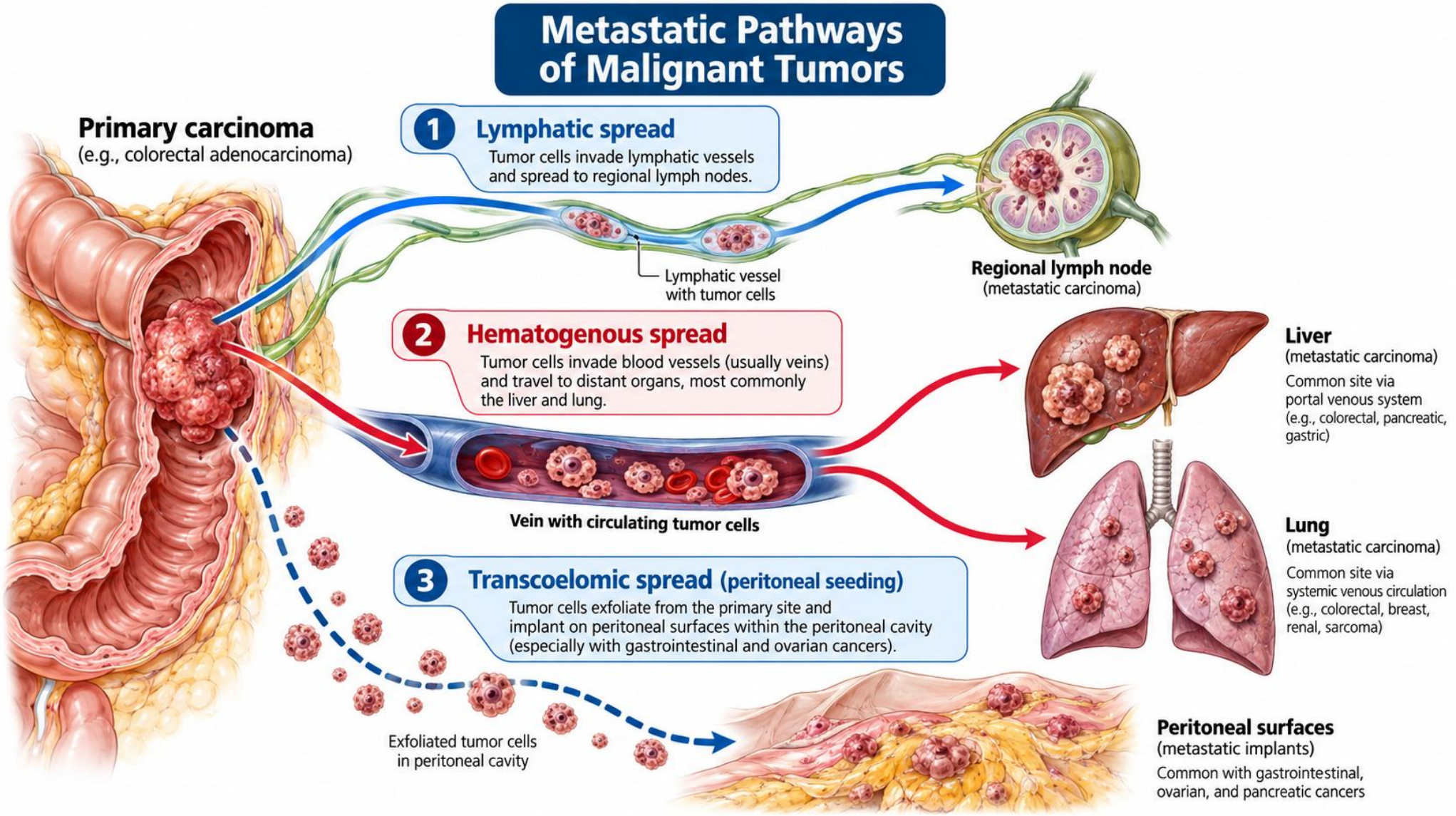
3-Attachment to novel ECM components .

4-Migration of tumor cells.

Ability to invade: ECM invasion in four steps



Metastatic cascade: dissemination and homing



Limitless Replicative Potential

- Most normal human cells have a capacity of 60 to 70 doublings.
- After this the cells lose the capacity to divide and enter senescence.
- This phenomenon is due to progressive shortening of telomeres at the ends of chromosomes.
- **Cancer cells** maintain telomeres and avoid senescence.
- Telomerase activation supports continued proliferation.
- This is a hallmark enabling long-term tumor growth.

Development of Sustained Angiogenesis:

- Tumors cannot enlarge beyond 1-2 mm in diameter unless they are vascularized.
- Cancer cells can stimulate neo-angiogenesis during which new vessels sprout from previously existing capillaries or in some cases vasculogenesis in which endothelial cells are recruited from the bone marrow.
- Angiogenesis is thus a necessary biologic correlate of neoplasia, both benign and malignant.
- Angiogenesis is required not only for continued tumor growth but also for access to the vasculature and hence for metastasis.

Reprogramming Energy Metabolism

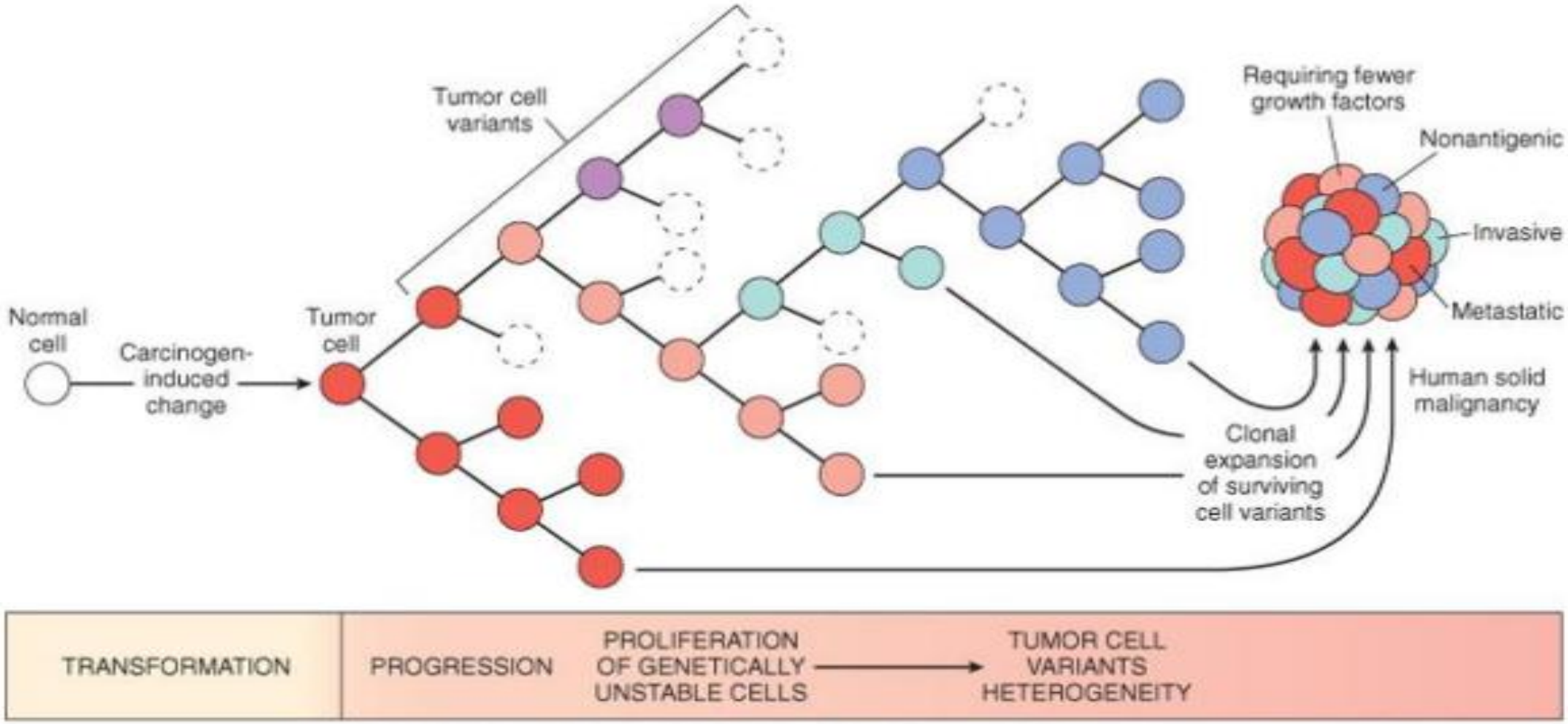
- Reprogramming of energy metabolism is so common to tumors that it is now considered a hallmark of cancer.
- Even in the presence of oxygen cancer cells shift their glucose metabolism from mitochondrial oxidative phosphorylation to glycolysis.
- This phenomenon, called the **Warburg effect** and also known as **aerobic glycolysis.**

Clinical relevance:

- Helps explain high glucose uptake by tumors.
- Forms the biologic basis for PET imaging.

Tumor progression:

- It is well established that over a period of time, many tumors become more aggressive and acquire greater malignant potential.
- This phenomenon is referred to as tumor progression.
- Tumor progression most likely result from multiple mutations that accumulate independently in different cells, generating subclones with different characteristics.



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Summary:

- Cancer progression results from accumulated mutations and clonal selection.
- Invasion requires detachment, ECM degradation, new attachment and migration.
- Metastasis requires vascular/lymphatic dissemination and colonization.
- Telomere maintenance supports limitless replication.
- Angiogenesis and the Warburg effect support growth and progression.

Thank you